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SCIENTIFIC ARTICLE

MRI radiomics for hamstring strain injury identification and return to sport classification: a pilot study

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Abstract

Objective To determine if MRI-based radiomics from hamstring muscles are related to injury and if the features could be used to perform a time to return to sport (RTS) classification. We hypothesize that radiomics from hamstring muscles, especially T2-weighted and diffusion tensor imaging-based features, are related to injury and can be used for RTS classification.

Subjects and methods MRI data from 32 athletes at the University of Wisconsin-Madison that sustained a hamstring strain injury were collected. Diffusion tensor imaging and T1- and T2-weighted images were processed, and diffusion maps were calculated. Radiomics features were extracted from the four hamstring muscles in each limb and for each MRI modality, individually. Feature selection was performed and multiple support vector classifiers were cross-validated to differentiate between involved and uninvolved limbs and perform binary ($\leq$ or $>$ 25 days) and multiclass ($<$ 14 vs. 14—42 vs. $>$ 42 days) classification of RTS.

Result The combination of radiomics features from all diffusion tensor imaging and T2-weighted images provided the most accurate differentiation between involved and uninvolved limbs (AUC $\approx 0.84 \pm 0.16$). For the binary RTS classification, the combination of all extracted radiomics offered the most accurate classification (AUC $\approx 0.95 \pm 0.15$). While for the multiclass RTS classification, the combination of features from all the diffusion tensor imaging maps provided the most accurate classification (weighted one vs. rest AUC $\approx 0.81 \pm 0.16$).

Conclusion This pilot study demonstrated that radiomics features from hamstring muscles are related to injury and have the potential to predict RTS.

Keywords Hamstring strain injury · Radiomics · Machine learning · Diffusion tensor imaging · Time to return to sport · Muscles

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Introduction

Hamstring strain injuries (HSIs) are prevalent in sprinting sports [1], such as American football [2], soccer, and track and field, and possess a burden on recreational and professional athletes alike. HSIs consist of damage to the hamstring muscle fibers and loss of functionality [1] and are characterized by persistent symptoms, such as posterior thigh pain, lengthy recovery periods, and high rates of reinjury [3, 4]. HSIs can result in time away from training and competition, reduction of athletic performance and team success, and financial loss [5].

Mixed evidence exists on the association between injury severity and the time needed for return to sport (RTS). However, it is speculated that increased injury severity is associated with an increased RTS time [6]. The current standard of care relies on clinicians to implement a battery of functional tests to determine RTS [7, 8]. However, 14–63% of athletes with an HSI, reinjure during the same season and up to two years after the initial injury [9]. Clinical evaluation/functional tests have been considered unreliable tools to predict RTS or reinjury [8, 10]; possibly because it does not take into consideration the time needed for natural healing of the muscle [7]. Thus, there has been increased scientific interest in refining predictions of RTS using quantitative imaging, such as magnetic resonance imaging (MRI), to improve the recovery process following HSIs and reduce the rate of reinjuries. Numerous studies have explored the potential of MRI data to predict clinical outcomes such as RTS [11–14] and the probability of re-injuring [9, 15], yet the findings are conflicting. Prior work investigating the ability of MRI to contribute to RTS predictions uses metrics from the gross anatomical level (i.e., volume, location, number of muscles involved) [16] but does not improve time to RTS predictions beyond physical examination and injury history [17, 18]. However, MRI-based injury grading scales, such as the British Athletics Muscle Injury Classification, do demonstrate a potential association with time to RTS [19–23]. Furthermore, moderate evidence suggests reinjury risk is associated with the hamstring muscle or the tissue-type involved in the index injury [15] while others have found that MRI descriptors of the index HSI do not accurately predict risk of reinjury [24].

Radiomics is a medical imaging analysis field that refers to the extraction of quantitative data from images that can be used during workups for clinical decision-making [25]. In radiomics, mathematically quantitative features are extracted from a region of interest, to capture possible areas of heterogeneity and then used to discover hidden data patterns [26]. Although radiomics was initiated in the field of oncology, it has the potential to be applied to other conditions [25]. Hence, the objective of this pilot study is to determine (1) if MRI-based radiomics from hamstring muscles are related to injury and (2) if the features could be used to perform both a binary and multiclass RTS classification. We hypothesize that radiomics from hamstring muscles, especially T2-weighted and diffusion tensor imaging-based features, are related to injury and can be used for RTS classification.

Subjects and methods

The data presented in this study were collected as part of a larger prospective cohort investigation of collegiate athletes who sustained an HSI. The study was approved by the University's Health Sciences Institutional Review Board, and participants provided written informed consent prior to enrollment. The proposed methodology includes the acquisition of MRI data, image preprocessing, muscle region of interest (ROI) segmentation, radiomics feature extraction, feature selection, and the implementation of a support vector classifier to perform classifications (Fig. 1a).

Study population

Division 1 collegiate athletes at the University of Wisconsin-Madison that sustained an HSI were recruited between 2017–2020. The following inclusion criteria were utilized for recruitment: division 1 athlete participating in football, track, or soccer; clinical diagnosis of acute HSI requiring ≥ 1-day time-loss from practice or competition, available for MRI scan at the time of injury (≤ 7 days from injury). Exclusion criteria included complete hamstring avulsion injury, bilateral injuries, MRI contraindications, and non-useable diffusion tensor imaging (DTI) data. Out of the 133 total injuries (51 completed MRI scans at the time of injury from 39 unique athletes), data from 32 subjects were included in this study (mean age = 19.56 ± 1.41 years, female = 7, see Table 1 for additional demographics information). Subjects with multiple injuries were included only once in the study, for the first injury with complete imaging. Four injuries were excluded as the DTI data was unusable due to incomplete fat saturation and movement artifacts, 2 were excluded due to bilateral injuries, and 1 was excluded due to missing data.

After the initial injury, a standardized rehabilitation protocol was implemented by the team's athletic trainer, and RTS was determined when medical clearance was obtained to resume all sport-related activities. RTS clearance was based on a combination of factors including full hamstring range of motion, minimal to no pain with hamstring palpation, and no apprehension with on-field sports-specific movements.

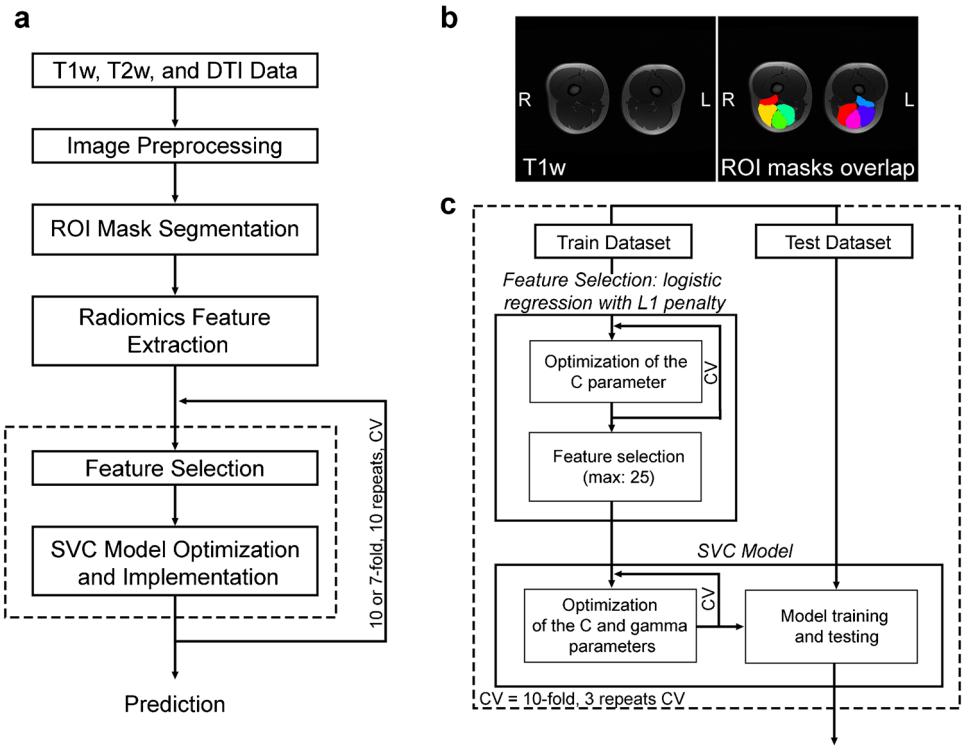

Fig. 1 Proposed methodology. **a** The main steps of the proposed methodology to apply radiomics to detect and assess hamstring strain injury are shown. T1-weighted (T1w), T2-weighted (T2w), and diffusion tensor imaging (DTI) data were obtained from collegiate athletes at the University of Wisconsin-Madison that sustained a hamstring strain injury. **b** Following preprocessing, the four hamstring muscles were delineated as regions of interest (ROIs) in each limb: biceps femoris-short head, biceps femoris-long head, semimembranosus, and semitendinosus (left (L) limb, colored light blue, blue, magenta, and red, respectively). Radiomics features were extracted from each ROI and image modality, individually. Extracted features were then used to train and test a support vector classifier (SVC) to differentiate between involved and uninvolved limbs and perform both a binary and multiclass classification of return to sport. **c** A cross-validation (CV) technique is implemented, and feature selection is performed using logistic regression with an L1 penalty. Grid search is used for the optimization of logistic regression and SVC's parameters

Table 1 Summarized demographic characteristics for the study population

Mean Age (years)	Sex	Sport	Involved Limb	RTS (days)	Primary Muscle Injured*	BAMIC
19.56 ± 1.41	Male = 25 Female = 7	Track = 17 Soccer = 1 Football = 14	Right = 18 Left = 14	32.13 ± 21.03	BFsh = 1 BFlh = 19 ST = 3 SM = 3	0a = 2 0b = 2 1a = 3 1b = 3 2a = 3 2b = 3 2c = 2 3a = 1 3b = 2 3c = 11

Involved = Injured, RTS = time to return to sport, BFsh = Biceps Femoris short head, BFlh = Biceps Femoris long head, ST = Semitendinosus, SM = Semimembranosus, BAMIC = British Athletic Muscle Injury Classification

*Primary injured muscle was only identified for an overall BAMIC grade > 0

Image acquisition, preprocessing, and region of interest segmentation

T1-weighted (T1w), T2-weighted (T2w), and DTI MRI images were obtained on a 3 T GE scanner (GE Healthcare Discovery MR750, Waukesha, WI) 32-channel torso coil (see Table 2 for acquisition parameters). Images were first converted to NIfTI format using the dcm2niix tool, version v1.0.20180622 [27]. For DTI images, FMRIB Software's FSL TOPUP and EDDY tools [28–30] were implemented for distortion, eddy current, and motion correction. Following correction, tensor fitting was performed to obtain maps of fractional anisotropy (FA), mean and radial diffusivity (MD and RD), and principal effective diffusivity eigenvalues (λ1 [axial diffusivity], λ2, λ3). Moreover, four ROIs were manually delineated, by selecting voxels representing muscle tissue of each respective hamstring muscle on the T1w images in each limb (FSLeyes, v1.5.0, Oxford, UK) [31] (Fig. 1b). ROIs were then interpolated across all slices to create a three-dimensional volume used to represent each hamstring muscle. Using the T2-weighted image, a musculoskeletal radiologist with over 20 years of experience identified the region of increased signal that was associated with an HSI. Manual segmentation was used to identify all voxels within this region (FSLeyes, v1.5.0, Oxford, UK) [31] and was further refined by taking the intersection of the muscle boundaries and the representative region. The resulting region was used to represent the region of injured muscle tissue. The ROIs were co-registered to the T2w images and DTI maps to obtain a version of each ROI mask spatially registered to each image space.

Table 2 MRI imaging acquisition parameters

Axial T1w	
time to echo	15.3 ms
repetition time	788 ms
acquisition matrix	1024 × 1024
number of slices	16
voxel size	0.43 × 0.43 × 9.00 mm
Axial T2w (Fat-suppressed weighted fat/water IDEAL)	
time to echo	88.6 ms
repetition time	4679 ms
acquisition matrix	512 × 512
number of slices	16
voxel size	0.86 × 0.86 × 9.00 mm
Axial DTI (Single shot echo planer repeated with reversed phase-encode direction)	
b	500 s/mm^2
directions	30
non-diffusion weighted volumes	6
time to echo	51.1 ms
repetition time	5770 ms
acquisition matrix	160 × 160
number of slices	72
slice thickness	3 mm
gap	0 mm
field of view	48 cm

T1w = T1-weighted, T2w = T2-weighted, DTI = diffusion tensor imaging, IDEAL = iterative decomposition of water and fat with echo asymmetry and least-squares sequence

Radiomics features extraction

Radiomics extraction was performed using the PyRadiomics v3.0.1 platform [32] a robust open-source Python package that calculates engineered features from medical images and is commonly used for the assessment of cancer lesions. Radiomics features were obtained for each MRI modality, individually (T2w, T1w, FA, MD, RD, λ1, λ2, λ3). Similarly, each MRI radiomics dataset individually contains features extracted from each hamstring muscle (as defined in the *Image Preprocessing and Region of Interest Segmentation* section). To determine the optimal bin width for image discretization, the minimum range of intensity values, within an ROI for each MRI modality, was identified and divided by 30. The obtained bin width ensures the utilization of at least 30 bins for the discretization of each image [33]. Since radiomics has not been previously applied for the assessment of HSI, there is no prior information regarding which features may be the best predictors. Therefore, all the features facilitated in PyRadiomics were extracted from the original images (no additional transformations/filters were applied). The following classes of radiomics were extracted: first-order statistics, shape-based, gray level cooccurrence matrix, gray level run length matrix, gray level size zone matrix, and gray level dependence matrix [32]. A total of 101 radiomics features were included in this study from each ROI (see Supplementary Table 1). Therefore, each MRI radiomics dataset contains a total of 404 (101 per ROI/muscle). The features were normalized by removing the mean and scaling to unit variance using the scikit-learn standard scaler function [34].

Feature selection and support vector classifier implementation

A support vector classifier (SVC) model was implemented to differentiate between involved and uninvolved limbs and for RTS classification using individual and combination of multimodal MRI radiomics features of HSIs. For the RTS classification, the problem was defined as both a binary and a multiclass one. A threshold of ≤ or > 25 days was used for the binary problem as it provides a 50/50 distribution of

subjects in each category ($n = 16$ for each group). For the multiclass classification, more clinically relevant thresholds were selected as < 14 days (2 weeks, $n = 7$), between 14 to 42 days (2–4 weeks, $n = 14$), and > 42 days (6 weeks, $n = 11$). SVC was implemented in this study due to its efficiency when processing high-dimensional datasets. SVC tends to perform well when working with small data sets when compared to more complex algorithms that may need larger datasets to prevent overfitting. Furthermore, to further prevent overfitting, feature selection was first performed to reduce the number of predictors input into the SVC (a maximum threshold of 25 features). Feature selection and SVC implementation were performed using Python and the scikit-learn 1.0.2 library [34].

To evaluate the performance of each SVC model, a 10- or 7-fold, 10 repeats, cross-validation evaluation technique was implemented, for the binary and multiclass classifications, respectively. A 7-fold CV was performed for the multiclass RTS classification, instead of a 10-fold, because the class with the least number of subjects has only 7. Thus, 7-fold cross-validation ensures that each category has at least one subject in the test dataset for model evaluation. Cross-validation was executed due to the small sample size of this study as there is no ideal data split (for training and testing) that would offer a satisfactory variance. Additionally, stratification was utilized to ensure the representation of each class during cross-validation. For each fold, the dataset was split into training and test datasets (training, $n = 29$, test, $n = 3$). The training dataset was used for feature selection and model training, while the test dataset was used only to evaluate the performance of the final model (Fig. 1c). A logistic regression model with L1 penalty, and *liblinear* solver, was implemented for feature selection on the training dataset within each cross-validation fold. Using a 10-fold, 3 repeats, cross-validation technique, a grid search optimization was implemented to obtain the optimal inverse of regularization strength from values between 0 and 2, with 0.01 increments. The inverse of regularization strength value was then used to train the logistic regression model. An optimal set of features was identified for each fold individually (maximum threshold – 25 features). Following feature selection, the optimal set of features identified by the logistic regression model was then used to train a SVC, with a radial basis function kernel, to perform the classification. The training dataset, containing the optimal set of features, and a 10-fold, 3 repeats, cross-validation grid search optimization, was used to find the optimal kernel coefficient, gamma (parameters options: 1, 0.1, 0.01, 0.001, 0.0001), and the regularization parameters (parameters options: 0.1, 1, 10, 100, 1000). The optimized kernel coefficient and regularization parameters were then applied to train the SVC classifier. Furthermore, the trained SVC was tested using the testing dataset; this feature selection and SVC implementation were performed for each cross-validation fold and repetition. The SVC performance was evaluated using the following metrics: mean test accuracy, the area under the receiver operating characteristic curve (AUC), sensitivity, specificity, weighted F1 score, precision, and recall, over each fold and repetition. A one vs. the rest (OvsR) classification method was implemented for the multiclass RTS classification, as SVC algorithms do not intrinsically support classification between multiple classes. Therefore, an OvsR approach was implemented in which the multiclass classification problem is split into multiple binary ones. Meaning that a SVC is trained on each binary classification (in this case (1) < 14 vs. [14 – 42, > 42] days, (2) 14—42 vs. [< 14, > 42] days, and (3) > 42 vs. [< 14, 14 – 42] days) and predictions are made using the most confident model. Additionally, since we have unbalanced classes, weighted metrics are used in this case (weighted OvsR). If no optimal set of features was found for a specific fold or repetition, no SVC model was trained, and the evaluation metrics for that fold were penalized by assigning a value of zero. This penalization would allow us to recognize models whose performance was most likely dominated by random chance, by significantly affecting their evaluation metrics (≤ 0.5).

Statistical analysis

Several evaluation metrics were implemented to evaluate the performance of the cross-validated SVCs. For the involved and uninvolved limbs and binary RTS classifications, the SVC performance was evaluated using mean test accuracy and AUC. AUC depends on the individual sensitivity and specificity values of each cross-validated model thus both are also shown. For the multiclass RTS classification, a weighted F1 score was implemented instead of accuracy since the latter does not take class imbalance into consideration which was present in our analysis. The F1 score depends on the individual precision and recall scores thus both were also presented for this case. Furthermore, to compare the performance of the multiclass RTS classifier with the binary counterpart, the F1 score was also calculated for the binary RTS classification. Additionally, 95% confidence intervals (CIs) were calculated for all the evaluation metrics using 1000 bootstrap resamples of the mean difference [35].

Statistical analysis was performed in Python using the SciPy library [36] to determine which MRI-based radiomics offered the most accurate classification for involved and uninvolved limb classification and RTS classification in HSI. A Friedman test was first used to test the null hypothesis that the mean test evaluation metric (accuracy, AUC, or F1 score) is equal for each SVC model trained with a different MRI-based radiomics input dataset [37–39]. If the Friedman test's null hypothesis was rejected, the evaluation metric of the model with the best performance was individually

compared to all the remaining models using a Wilcoxon signed-rank test [40]. Since the Wilcoxon signed-rank test was performed multiple times for each evaluation metric, Bonferroni was implemented for the correction of multiple comparisons. Therefore, the individual *p-values* obtained from the Wilcoxon signed-rank tests when analyzing each evaluation metric were corrected using the *multipletests* function, part of the statsmodels Python module, using an alpha value (family-wise error rate) of 0.05.

Results

Radiomics features are related to hamstring strain injury

To determine if radiomics features from hamstring muscles in collegiate athletes with HSI are related to injury, an automatic feature selection (≤ 25 features) and SVC were implemented to differentiate between involved and uninvolved limbs. The highest mean test accuracy and AUC are obtained when selecting the optimal set of features from all DTI maps and T2w combined (DTI-T2w, Fig. 2). A value of approximately 0.84 (± 0.16) was obtained for both mean test accuracy and AUC and it was statistically significantly higher than using individual MRI radiomics datasets. However, using DTI-T2w radiomics features does not provide statistically significantly higher performance than the combination of all the DTI maps-based features or DTI, T2w, and T1w features. Furthermore, when evaluating the classification performance using individual MRI radiomics, MD and T2-based features perform better. The lowest classification performance was obtained when selecting the optimal set of features from FA, λ1, or T1w-based radiomics, in which no optimal features are selected for 20, 29, and 47 out of the 100 cross-validation repetitions, respectively. As a result, the classification performance for these cases was dominated by random chance and the absence of an optimal set of features only occurred in these cases. Table 3 shows the specific mean test accuracy, AUC, sensitivity, and specificity, values for each cross-validated SVC model, along with their respective standard deviations and 95% CIs. Supplementary Fig. 1 shows the ROCs for all the cross-validated SVCs. Additionally, the supplementary section *Selected features for involved vs. uninvolved limb classification* shows the features selected during the cross-validation of the multiple SVCs. A total of 43 features were selected to be part of the optimal set (≤ 25 predictors) at least once during feature selection. During cross-validation, the T2w-derived first-order skewness and second-order small area emphasis, gray level size zone matrix, features were selected 100% and 98% of the time, respectively.

Radiomics features to classify return to sport: a binary classification problem

An automatic feature selection (≤ 25 features) and SVC were implemented to classify between subjects with an RTS ≤ or > 25 days (16 subjects in each category, $n = 32$). The most accurate RTS binary classification was obtained when selecting the optimal set of features from all extracted radiomics (DTI maps, T2w, and T1w) (Fig. 3). A value of approximately 0.95 (± 0.15) was obtained for both mean test accuracy and AUC. These metrics

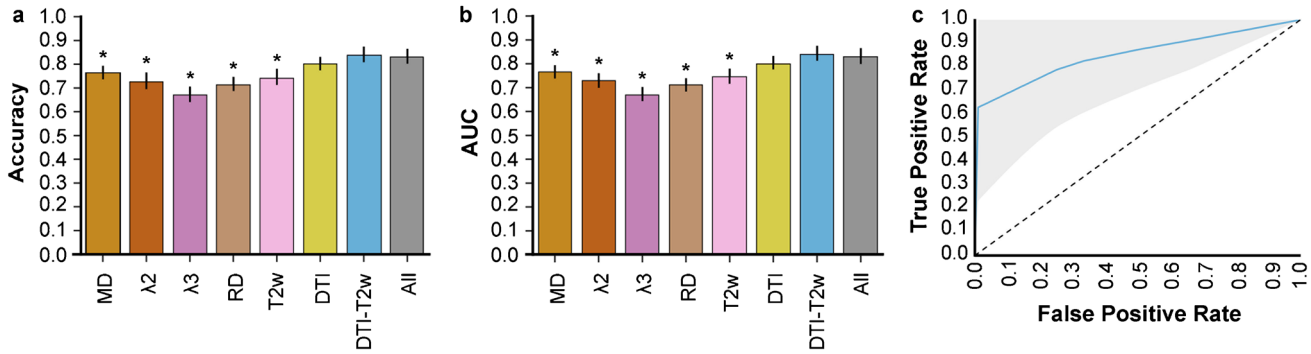

Fig. 2 Radiomics features are related to hamstring strain injury**.** Feature selection and a support vector classifier (SVC) were implemented using a 10-fold, 10 repeats, cross-validation technique to differentiate between involved and uninvolved limbs. Eleven SVCs were cross-validated using individual and a combination of MRI-based radiomics datasets as input. **a** Mean test accuracy and **(b)** area under the receiver operating characteristic (ROC) curve (AUC) are shown with their respective 95% confidence intervals (black line). DTI = combination of FA, MD, λ1, λ2, λ3, and RD features, DTI-T2w = combination of all DTI and T2w features, and All = combination of all DTI, T2w, and T1w features. *Denotes statistically significant difference from DTI-T2w (adjusted *p-value* < 0.05). **c** The ROC curve for the best predictor, the DTI-T2w radiomics dataset, is shown with mean test AUC standard deviation represented by the gray-highlighted area. The black dotted line indicates the ROC threshold under which the performance of the model is dominated by random chance. **a-b** Results for FA, λ1, and T1w-based radiomics are not shown since their performance was dominated by random chance

Table 3 Results for involved vs. uninvolved limb diffrentiation and binary return to sport classification

Radiomics Dataset	Accuracy	AUC	Sensitivity	Specificity	Weighted F1 Score
Involved vs. Uninvolved Classification (10-fold, 10 repeats, cross-validation)					
MD	0.764 (±0.147) (0.736, 0.794)	0.767 (±0.148) (0.739, 0.795)	0.738 (±0.256) (0.685, 0.789)	0.796 (±0.214) (0.751, 0.841)	-
λ2	0.730 (±0.166) (0.696, 0.764)	0.731 (±0.166) (0.700, 0.761)	0.633 (±0.254) (0.583, 0.681)	0.829 (±0.212) (0.787, 0.871)	-
λ3	0.671 (±0.160) (0.641, 0.703)	0.673 (±0.158) (0.643, 0.703)	0.555 (±0.264) (0.501, 0.602)	0.791 (±0.233) (0.746, 0.838)	-
RD	0.716 (±0.145) (0.687, 0.747)	0.714 (±0.144) (0.684, 0.740)	0.629 (±0.252) (0.582, 0.680)	0.798 (±0.236) (0.752, 0.846)	-
T2w	0.746 (±0.166) (0.712, 0.781)	0.748 (±0.164) (0.717, 0.780)	0.677 (±0.267) (0.626, 0.729)	0.818 (±0.227) (0.776, 0.864)	-
DTI	0.804 (±0.141) (0.776, 0.831)	0.804 (±0.142) (0.776, 0.833)	0.818 (±0.207) (0.779, 0.857)	0.790 (±0.217) (0.749, 0.830)	-
DTI-T2w	0.842 (±0.163) (0.809, 0.875)	0.844 (±0.163) (0.813, 0.875)	0.804 (±0.233) (0.757, 0.850)	0.884 (±0.193) (0.843, 0.918)	-
All	0.834 (±0.160) (0.803, 0.865)	0.834 (±0.161) (0.800, 0.865)	0.793 (±0.233) (0.746, 0.838)	0.875 (±0.185) (0.840, 0.910)	-
RTS Binary Classification (≤ 25 days vs. ≥ 25 days, 10-fold, 10 repeats, cross-validation)					
FA	0.881 (±0.208) (0.841, 0.921)	0.885 (±0.213) (0.838, 0.925)	0.900 (±0.245) (0.855, 0.945)	0.870 (±0.270) (0.815, 0.920)	0.876 (±0.216) (0.835, 0.918)
MD	0.719 (±0.273) (0.665, 0.772)	0.720 (±0.281) (0.663, 0.778)	0.740 (±0.384) (0.655, 0.820)	0.700 (±0.406) (0.625, 0.780)	0.684 (±0.302) (0.621, 0.740)
λ1	0.724 (±0.269) (0.674, 0.778)	0.730 (±0.280) (0.673, 0.783)	0.760 (±0.371) (0.680, 0.830)	0.700 (±0.374) (0.625, 0.770)	0.702 (±0.288) (0.647, 0.754)
λ2	0.736 (±0.233) (0.691, 0.782)	0.730(±0.252) (0.686, 0.780)	0.735 (±0.390) (0.660, 0.810)	0.725 (±0.363) (0.650, 0.795)	0.704 (±0.257) (0.654, 0.753)
λ3	0.863 (±0.219) (0.820, 0.905)	0.868 (±0.222) (0.825, 0.905)	0.865 (±0.282) (0.805, 0.915)	0.870 (±0.288) (0.805, 0.925)	0.853 (±0.237) (0.806, 0.895)
RD	0.813 (±0.248) (0.766, 0.861)	0.808 (±0.260) (0.755, 0.860)	0.835 (±0.340) (0.765, 0.895)	0.780 (±0.356) (0.710, 0.855)	0.789 (±0.277) (0.731, 0.842)
DTI	0.933 (±0.181) (0.895, 0.966)	0.933 (±0.187) (0.895, 0.970)	0.945 (±0.199) (0.905, 0.980)	0.920 (±0.242) (0.870, 0.965)	0.928 (±0.189) (0.892, 0.962)
DTI-T2w	0.941 (±0.175) (0.905, 0.973)	0.938 (±0.185) (0.900, 0.970)	0.935 (±0.220) (0.885, 0.970)	0.940 (±0.215) (0.900, 0.980)	0.936 (±0.184) (0.900, 0.969)
All	0.953 (±0.158) (0.918, 0.980)	0.958 (±0.150) (0.928, 0.983)	0.960 (±0.183) (0.920, 0.990)	0.955 (±0.175) (0.915, 0.985)	0.950 (±0.171) (0.915, 0.981)

Each mean test metric value is shown with their respective standard deviation and 95% confidence interval. RTS = time to return to sport, FA = fractional anisotropy, MD = mean diffusivity, RD = radial diffusivity, λ1, λ2, and λ3 = principal effective diffusivity eigenvalues, T2w = T2-weighted, T1w = T1-weighted, DTI = combination of FA, MD, λ1, λ2, λ3, and RD, DTI-T2w = combination of DTI and T2w, All = combination of DTI, T2w, and T1w, AUC = area under the receiver operating characteristic curve. For the involved vs. uninvolved classification, results for FA, λ1, and T1w radiomics are not shown since their performance was dominated by random chance. Likewise, for the binary RTS classification, the results for T1w and T2w radiomics are not displayed

are statistically significantly higher than performing the classification using radiomics features extracted from an MRI modality individually, but not than using the combination of all the DTI maps-based features or DTI and T2w features. Moreover, FA-based radiomics features offer the most accurate RTS binary when assessing the performance of individual MRI-based radiomics datasets. The lowest classification performance was obtained when selecting the optimal set of features from T2w and T1w-based radiomics, in which no optimal features are selected for 64 and 38 out of the 100 cross-validation repetitions, respectively. As a result, the RTS binary classification performance for these cases was dominated by random chance. The absence of an optimal set of features occurred ≤ 2% of the time for

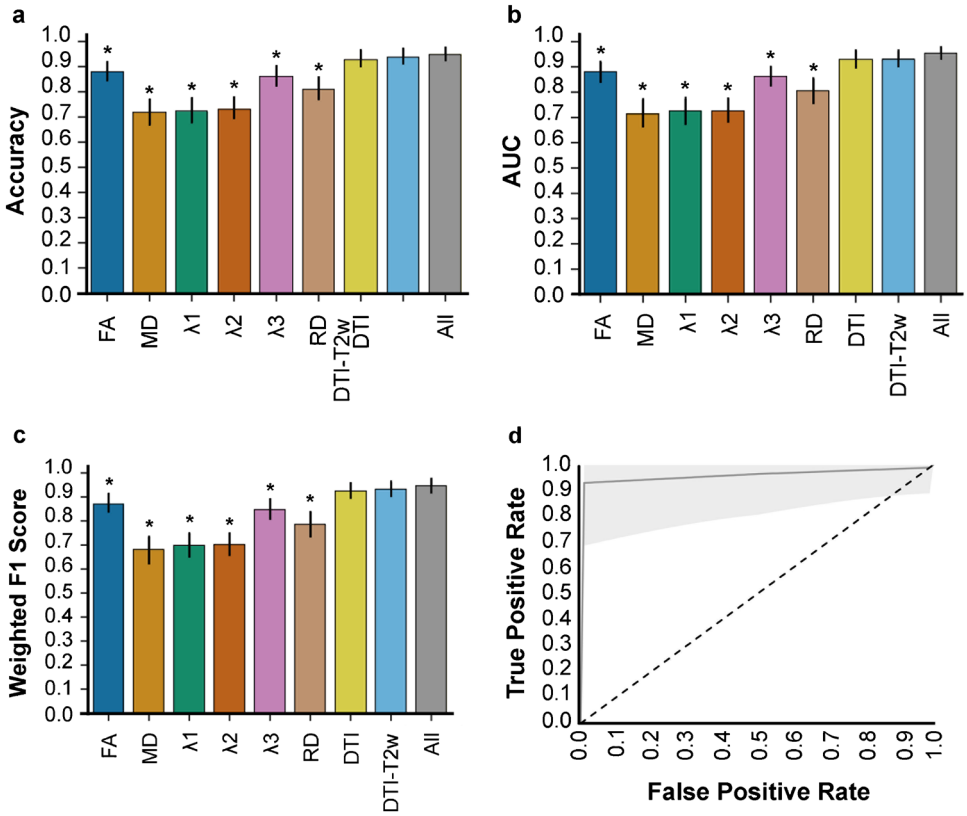

Fig. 3 Radiomics features predict time to return to sport: a binary classification problem. Feature selection and a support vector classifier (SVC) were implemented using a 10-fold, 10 repeats, cross-validation technique to perform a binary classification of time to return to sport (RTS). Subjects were classified into two groups: RTS ≤ 25 ($n=16$) or > 25 days ($n=16$). Eleven SVCs were cross-validated using individual and a combination of MRI-based radiomics datasets as input. **a** Mean test accuracy, **b** area under the receiver operating characteristic (ROC) curve (AUC), and **(c)** weighted F1 score are shown with their respective 95% confidence intervals (black line). DTI = combination of FA, MD, λ1, λ2, λ3, and RD features, DTI-T2w = combination of all DTI and T2w features, and All = combination of all DTI, T2w, and T1w features. *Denotes statistically significant difference from All (adjusted *p-value* < 0.05). **d** The ROC curve for the best predictor, the All radiomics dataset, is shown with mean test AUC standard deviation represented by the gray-highlighted area. The black dotted line indicates the ROC threshold under which the performance of the model is dominated by random chance. **(a-c)** Results for T1w and T2w-based radiomics are not shown since their performance was dominated by random chance

the remaining input datasets. Table 3 shows the specific mean test accuracy, AUC, sensitivity, and specificity, for each one of the cross-validated SVC models, along with its respective standard deviations and 95% CIs. Supplementary Fig. 2 shows the ROCs for all the cross-validated SVCs. Additionally, the supplementary section *Selected features for binary RTS classification* shows the features selected during the cross-validation of the multiple SVCs. A total of 54 features were selected to be part of the optimal set (≤ 25 predictors) at least once during feature selection. During cross-validation, the FA-derived first-order 10 percentile, 90 percentile, and kurtosis features were selected 100%, 99%, and 99% of the time, respectively. Additionally, the λ2-derived second-order cluster prominence, gray-level cooccurrence matrix feature was selected 96% of the time.

Radiomics features to classify time to return to sport: a multiclass classification problem

We expanded the RTS classification from a binary problem to a multiclass one (< 14 vs. 14—42 vs. > 42 days). The most accurate RTS multiclass classification was obtained when selecting the optimal set of features (≤ 25 predictors) from all DTI maps (Fig. 4). A value of approximately 0.81 (± 0.16) was obtained for weighted OvsR AUC and 0.93 (± 0.19) for weighted F1 score (Table 4). These metrics are statistically significantly higher than performing the classification using individual MRI radiomics datasets, except λ3. DTI-based radiomics do not offer statistically significantly higher performance than the combination of DTI and T2w-based features or DTI, T2w, and T1w. The lowest classification performance was obtained when selecting

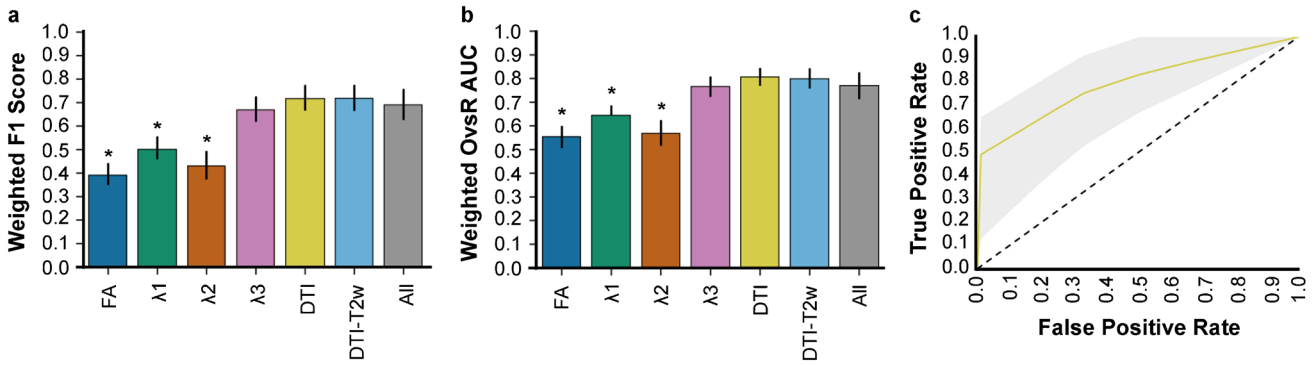

Fig. 4 Radiomics features predict time to return to sport: a multiclass classification problem. Feature selection and a support vector classifier (SVC) were implemented using a 7-fold, 10 repeats, cross-validation technique to perform a multiclass classification of time to return to sport (RTS). Subjects were classified into three groups: RTS < 14 ($n=7$), between 14 to 42 ($n=14$), and > 42 days ($n=11$). Eleven SVCs were cross-validated using individual and a combination of MRI-based radiomics datasets as input. **a** Mean test weighted area under the receiver operating characteristic (ROC) curve (AUC) and **(b)** F1 score are shown with their respective 95% confidence intervals (black line). DTI = combination of FA, MD, λ1, λ2, λ3, and RD features, DTI-T2w = combination of all DTI and T2w features, and All = combination of all DTI, T2w, and T1w features. *Denotes statistically significant difference from DTI (adjusted *p-value* < 0.05). **c** The weighted ROC curve for the best predictor, the DTI radiomics dataset, is shown with the mean test weighted AUC standard deviation represented by the gray-highlighted area. The black dotted line indicates the ROC threshold under which the performance of the model is dominated by random chance. **a-b** Results for MD, RD, T1w, and T2w-based radiomics are not shown since their performance was dominated by random chance

Table 4 Results for multiclass return to sport classification

Radiomics Dataset	Weighted F1 Score	Weighted OvsR AUC	Weighted Precision	Weighted Recall
	RTS Multiclass Classification (< 14 vs. 14–42 vs. 42 days, 7-fold, 10 repeats, cross-validation)			
FA	0.393 (±0.205) (0.350, 0.443)	0.556 (±0.197) (0.507, 0.599)	0.373 (±0.223) (0.319, 0.427)	0.471 (±0.210) (0.421, 0.519)
λ1	0.505 (±0.211) (0.458, 0.557)	0.647 (±0.167) (0.607, 0.686)	0.509 (±0.261) (0.449, 0.572)	0.573 (±0.181) (0.532, 0.615)
λ2	0.432 (±0.259) (0.374, 0.495)	0.571 (±0.241) (0.517, 0.624)	0.427 (±0.292) (0.359, 0.492)	0.487 (±0.253) (0.428, 0.549)
λ3	0.673 (±0.223) (0.618, 0.727)	0.769 (±0.191) (0.723, 0.809)	0.683 (±0.230) (0.631, 0.737)	0.712 (±0.220) (0.663, 0.764)
DTI	0.721 (±0.227) (0.666, 0.776)	0.809 (±0.162) (0.770, 0.844)	0.729 (±0.244) (0.676, 0.783)	0.761 (±0.203) (0.716, 0.808)
DTI-T2w	0.721 (±0.235) (0.665,0.776)	0.801 (±0.185) (0.758, 0.841)	0.740 (±0.250) 0.681, 0.795)	0.759 (±0.214) (0.711, 0.808)
All	0.694 (±0.283) (0.628, 0.759)	0.772 (±0.250) (0.713, 0.827)	0.703 (±0.292) (0.632, 0.767)	0.733 (±0.269) (0.669, 0.796)

Each mean test metric value is shown with their respective standard deviation and 95% confidence interval. RTS = time to return to sport, FA = fractional anisotropy, MD = mean diffusivity, RD = radial diffusivity, λ1, λ2, and λ3 = principal effective diffusivity eigenvalues, T2w = T2-weighted, T1w = T1-weighted, DTI = combination of FA, MD, λ1, λ2, λ3, and RD, DTI-T2w = combination of DTI and T2w, All = combination of DTI, T2w, and T1w, OvsR = One vs. the Rest, AUC = area under the receiver operating characteristic curve. Results for MD, RD, T2w, and T1w radiomics are not shown since their performance was dominated by random chance

the optimal set of features from MD, RD, T2w, and T1w-based radiomics, in which no optimal features are selected for 15, 14, 20, and 65 out of the 70 cross-validation repetitions, respectively. Consequently, the RTS multiclass classification performance for these cases was dominated by random chance. The absence of an optimal set of features occurred < 9% of the time for the remaining input datasets. Table 4 shows the specific weighted F1 score, OvsR AUC, precision, and recall for each one of the cross-validated SVC models, along with its respective standard deviations and

95% CIs. Supplementary Fig. 3 shows the ROCs for all the cross-validated SVCs. Additionally, the supplementary section *Selected features for multiclass RTS classification* shows the features selected during the cross-validation of the multiple SVCs. A total of 44 features were selected to be part of the optimal set (≤ 25 predictors) at least once during feature selection. During cross-validation, the FA-derived first-order 90 percentile and λ3-derived first-order 90 percentile and second-order size zone non-uniformity normalized, gray level size zone matrix, features were selected 100%, 96%, and 96% of the time, respectively.

Discussion

HSIs are predominant in high-speed running sports and are characterized by persistent pain and a high rate of reinjury [3, 4]. Premature RTS may be one possible reason for an increased chance of reinjury [7, 9]. Clinicians use functional sport-specific tests to decide on an RTS length, but these have been shown to underestimate the extent of injury in many cases [7, 8, 10]. To decrease the rate of reinjury, MRI exams are becoming more routinely employed to assess HSI and potentially find correlations between imaging parameters and RTS [11–14]. However, the potential of using MRI for RTS prediction is still unclear. Therefore, this pilot study establishes the potential of multimodal MRI radiomics analysis to assess HSIs. Radiomics analysis was first created to assess cancer lesions by extracting mathematically quantitative features from a region of interest and using them to model data patterns that would improve the disease assessment [26]. Additionally, its potential as an assessment tool for other clinical conditions has been previously discussed [25]. In this study, we utilized multimodal MRI data from college athletes that sustained a HSI to determine if radiomics features from hamstring muscles are related to injury and have the potential to predict time to return to sport. Our results revealed that the combination of DTI and T2w radiomics features are the strongest predictors when differentiating between involved and uninvolved limbs. Furthermore, the potential of radiomics for RTS prediction in HSI was revealed by means of both a binary and multiclass classification. To this end, the combination of radiomics features extracted from all MRI multimodalities (DTI, T1w, and T2w) and DTI provided the most accurate RTS classification, respectively.

A differentiation between involved and uninvolved limbs provides confidence that extracted radiomics features are related to injury. Based on a preliminary statistical analysis, it was hypothesized that radiomics features from DTI maps and T2w would be predictors of HSI [41]. To this end, it was found that the combination of radiomics features extracted from all DTI and T2w images offers the most accurate classification; with T2w and MD features as strong predictors when used individually. When analyzing which DT1-T2w features had the greater impact on the classification results, T2w-derived first-order skewness and second-order small area emphasis features were the only ones selected as part of the optimal set more than 90% of the time. Skewness is a measurement of the asymmetry of the distribution of pixel values about the mean. While small area emphasis is indicative of smaller zones and finer textures. This finding is not unexpected as the diagnosis of HSI is preferably supported by T2w imaging which is sensitive to fiber disruption and muscle edema [18, 42] and there is a 28/32 incidence of edema confirmed by a radiologist in our cohort. Therefore, the results suggest that these radiomics features may be relying upon changes in the muscle caused by edema to detect injury. However, findings from this study may also carry particular importance for athletes reporting clinical signs of HSI and inability to participate but no incidence of edema on a T2w image (4/32 in our cohort). Likewise, the application of DTI to detect subclinical signs of muscle injury has shown promising results since it is sensitive to changes in tissue microstructure [41, 43, 44]. Particularly, changes in DTI are associated with variations in water diffusion [45], which may be related to underlying pathologies such as edema or the disruption of diffusivity barriers due to muscle injury [43]. Besides the two T2w-derived features mentioned above, the MD-derived second-order size zone non-uniformity normalized feature was selected 80% of the time as part of the optimal set. The size zone non-uniformity normalized measures the variability of pixel values in the image (higher values indicate more heterogeneity), thus suggesting this radiomics feature may be sensitive to microstructural changes in the muscles.

MRI-based radiomics features from muscles, allow RTS classification in collegiate athletes with HSIs. For this study, RTS classification was first defined as a binary problem and a 25-day RTS cutoff was selected to provide a 50/50 distribution of subjects in each category. While the use of a continuous RTS prediction would be ideal, the use of a 25-day cutoff threshold for this study may still hold clinical value as the average RTS reported in the literature varies from 11 to 25 days [46, 47]. Our results reveal that radiomics features from DTI images pose a potential to predict RTS, with FA, λ3, and RD being the strongest individual predictors. However, the classification accuracy can be significantly improved by combining all the DTI features and/or adding T2w, and T1w features. The top features within the optimal set were DTI-derived features for both the binary and multiclass RTS classifications. The literature regarding the potential of MRI to predict clinical outcomes such as RTS is conflicting [11–14]. However, our results seem to align with previous literature when focusing on T1w and T2w imaging, since these radiomics features are not strong predictors of RTS individually but appear to aid the classification when combined with DTI-based descriptors. Furthermore, when performing a multiclass classification of

RTS the overall classification performance decays compared to the binary counterpart; these results are expected since the number of subjects per category is much lower.

While we have demonstrated the potential of DTI and T2w radiomics to detect HSIs and for RTS classification, there are some limitations to this study. Our study is susceptible to overfitting due to the small sample size and the absence of a separate test dataset. A cross-validation approach and feature selection, per fold and repetition, were applied to reduce overfitting and give a conservative estimate of model performance. A previous study showed that the number of radiomics features must be less than the number of subjects in the study to prevent overfitting [48]. Therefore, during feature selection, we reduced the number of radiomics predictors input into the SVC to ≤ 25. Although we recognize the limitation of our sample size, this study utilizes a clinically relevant and heterogenous MRI dataset of HSIs in collegiate athletes. Most importantly, it emphasizes the importance of generating larger and more robust datasets to fully explore the potential of radiomics as a clinical assessment tool of HSI. Furthermore, the registered RTS for collegiate and professional athletes tends to be premature and not always optimal to decrease the chance of reinjury [7, 9]; as athletes are expected to RTS as soon as possible. Thus, longitudinal data are needed to ensure an optimal RTS ground truth. Additionally, the use of a cutoff to identify those with RTS before or after 25-days is a limitation of this study. An ideal RTS prediction would be continuous, but a regression model would require a significant increase in sample size to guarantee acceptable performance. Lastly, RTS prediction by clinical evaluation was not included in this study for comparison. Despite these limitations, the strong performance of these models, combined with the uniqueness of the cohort data, provides confidence in the applicability of radiomics to assess HSIs and classify RTS.

In summary, we investigated the applicability of radiomics from multimodal MRI data as an assessment tool for HSIs. It was determined that radiomics features from hamstring muscles are related to injury and have the potential to predict RTS. Radiomics from DTI maps appear to be the strongest predictors suggesting these features may be capturing some of the underlying muscle microstructural changes due to injury. While T2w-derived features may be sensitive to changes in the muscle caused by edema. Taken together, this pilot study concludes that radiomics analysis on MRI data, specifically DTI, could support the assessment of musculoskeletal injuries such as HSIs and that larger and more robust longitudinal datasets are needed to fully understand its potential.

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Data availability The data in support of the findings of this study is not publicly available and cannot be shared due to Institutional Review Board's restrictions.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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